



Society of Petroleum Engineers

SPE-229934-MS

Carbon Nanocomposite Pickering Emulsions: A Promising Approach for Waterflood Conformance Control in Malaysian Reservoirs

Muhammad Mohsin Yousufi, Iskandar bin Dzulkarnain, and Mysara Eissa Mohyaldinn Elhaj, Department of Petroleum Engineering, Universiti Teknologi Petronas, Malaysia; Shehzad Ahmed, Chemical and Petroleum Engineering dept, Khalifa University, currently in CSIRO Energy, Kensington

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229934-MS](https://doi.org/10.2118/229934-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Objectives/Scope: Low sweep efficiency during water injection hinders oil production in mature Malaysian fields undergoing waterflood operations. This water conformance issue can lead to premature breakthrough in heterogeneous reservoirs, resulting in low oil recovery. This study details comprehensive experimental work evaluating Pickering particle-stabilized emulsions as conformance control agents, considering offshore Malaysian reservoir characteristics. Pickering emulsions offer superior stability against mechanical and thermal changes during reservoir propagation compared to surfactant-based emulsions. Furthermore, their rheology allows for lower pumping requirements and minimizes formation damage compared to foam and gel, enabling better selective placement and deeper reservoir penetration under high temperature, salinity, and pressure. For conformance control, these particle-based emulsions provide enhanced pore throat plugging in high-permeability thief zones, primarily due to improved adsorption of Pickering particles onto the rock surface.

Methods, Procedures, Process: This study explored an ecofriendly, novel carbon nanocomposite-based Pickering emulsion for conformance control in the high-temperature and salinity environments typical of Malaysian oil reservoirs. The research assessed the emulsion's thermal stability and rheological behavior under varying temperatures and salinities. Rock-fluid interaction, specifically the emulsion's adsorption on sandstone, was analyzed using contact angle measurements. Core flooding tests were conducted to evaluate plugging performance by quantifying permeability reduction under conditions closely simulating the reservoir.

Results, Observations, Conclusions: The study demonstrated highly effective plugging, with a substantial permeability reduction of 98.2% observed under dynamic conditions, while maintaining colloidal stability.

Novel/Additive Information: The insights from this study provide a basis for optimizing emulsion design early in the development process, leading to a more effective conformance control agent for reservoir applications. Importantly, the newly developed, ecofriendly emulsion, owing to its low volume consumption

and superior plugging performance, promises to significantly enhance conformance control performance in Malaysian reservoirs by mitigating excess water production and consumption, thereby reducing operational costs and unlocking oil production from previously untapped low-permeability zones.

Introduction

In the current oil industry climate, maximizing hydrocarbon recovery from mature wells is crucial to meet global energy demand while transitioning to sustainable sources (IEA World Energy Outlook 2019). One effective strategy involves improving reservoir profile by reducing heterogeneity. This is often achieved by plugging high-permeability zones to divert water flow towards low-permeability areas containing significant oil reserves (Fink 2021). While polymer gels have been widely used for this purpose, their drawbacks, such as high pumping requirements and potential formation damage, have limited their effectiveness (X. Zhang et al. 2022; S. Li et al. 2020).

Emulsions, known for their superior rheological properties and stability, offer a promising alternative to polymer (Patel et al. 2021; Adewunmi et al. 2022; Sun et al. 2022; Mandal et al. 2012). They can be injected at lower flow rates, reducing formation damage and improving sweep efficiency (Fu et al. 2022; Bai et al. 2000). Unlike polymer gels, emulsions can be selectively placed in deeper zones, enhancing oil recovery. While emulsions have been used in various oilfield applications, their use in conformance control has been relatively limited (Bai et al. 2000; McAuliffe 1973).

However, traditional surfactant-based emulsions can be environmentally and economically challenging due to their high emulsifier concentrations and potential toxicity. Developing more sustainable and efficient emulsion formulations is essential to fully realize their potential in improving oil recovery and reducing environmental impact.

Pickering emulsions, stabilized by solid particles rather than surfactants, offer several advantages. They require lower emulsifier concentrations, are often biodegradable and non-toxic, and can be tailored for various applications based on particle properties like conductivity, porosity, and luminosity (W. Li et al. 2022; Hossain, Deeming, and Edler 2021; Yang et al. 2017; Sarkar et al. 2019). These emulsions are highly stable due to the steric barrier created by the particles at the oil-water interface, preventing droplet coalescence and flocculation. This stability is attributed to the strong interfacial forces between the particles and the droplets, making them difficult to dislodge (Du Sorbier, Aimable, and Pagnoux 2015). As a result, Pickering emulsions provide superior protection against droplet aggregation compared to traditional surfactant-based emulsions.

While Pickering emulsions are generally stable against coalescence, they can experience creaming in high-salinity conditions. This occurs due to the density difference between the oil and brine phases, which disrupts the ionic balance surrounding the droplets. The excess ions introduced by the high salinity can shrink the electric double layer, reducing the electrostatic repulsion between droplets and leading to creaming. To address this instability, Pickering emulsions often require either functionalization of the particles for generating adhesiveness for both oil and water phases or inclusion of additional components, such as surfactants, excess particles, or polymers, to maintain their stability in high-salinity environments. Although effective, these techniques tend to increase the cost, complexity and toxicity concerns. So far hydrophilic silica, sodium montmorillonite, fly ash, chitin, and cellulose nanocrystals have been tested but have shown limited ability to stabilize oil-water interfaces on their own and require the assistance of surfactants or are functionalized to achieve the desired stability (S. Li et al. 2020; Xu et al. 2017; Kim et al. 2017; Wang et al. 2022; Pandey et al. 2018; Hagar et al. 2023).

Materials with high adsorption energy, particularly porous particles hold potential. These porous structures possess large surface area providing significant space for interaction and entrapment of oil and water molecules with minimal emulsifier usage (Frijters, Gunther, and Harting 2012; Y. Zhang et al. 2023). The low density and high adsorption capacity allows finer droplet dispersion. Currently mesoporous silica,

zeolite, and metal organic frameworks (MOFs) have been used as emulsifiers for different applications (Zeng et al. 2024; Abu Zaid et al. 2024; Shi et al. 2020), but these are relatively expensive.

Considering these aspects in this study a nanoporous material was designed through a hydrothermal process. The material incorporates bentonite and activated carbon, which are both porous materials with complementary hydrophilic and lipophilic characteristics respectively (Saafie et al. 2023). The hydrothermal reaction occurs between the two materials mixed in water in an autoclave under high temperature (200°C) and pressure conditions for 24 hours. During this nucleation and growth process particles break further into smaller components and interact with one another at an atomic level to form a new material (Wu et al. 2022). The resulting nanocomposite was used as an emulsifier and characterized for its stability and compatibility as a viable conformance agent. Its thermal and shear stability were evaluated, and its compatibility with sandstone was assessed using contact angle measurements under varying temperature, and salinity conditions. Finally, core flooding experiments were conducted to evaluate the nanocomposite's ability to plug sandstone formations.

Methodology

Materials

Activated carbon, bentonite and sodium chloride were purchased from Merck. Deionized water was used as the aqueous phase; Diesel oil was purchased from local petrol station. Miri Light Crude Oil, (API gravity = 37) was used for saturating rock pellets. Berea sandstone cores were purchased from Polygon Synergy Ventures Sdn Bhd.

Synthesis

The oil-in-water emulsion having 30:70 oil: water ratio for optimized viscosity to mobility performance was synthesized. The nanocomposite emulsifier was mixed with deionized water for 10 minutes to form a micelle state at moderate speed. This is followed by atomic dispersion of oil phase into the micelle with mixing speed increasing corresponding to the increase in volume and viscosity of the mixture. Throughout the synthesis temperature is set above boiling point of water (110°C) to prevent water evaporation courtesy of the Leidenfrost effect (Khan et al. 2014). This is because in such a condition the multiple Leidenfrost points due to fast reaction kinetics (such as water vapors, emulsifier, oil droplets) act as barriers to prevent the water from escaping and creating a separate phase. hereby creating an emulsion that is insensitive to high temperatures which is much needed under reservoir conditions. Furthermore, in some cases the high thermal energy causes the particles to gain higher zeta potential due to excitation of the ionic charges, this can enhance the interparticle repulsion mechanism leading to a more stable emulsion (Yousufi et al. 2023). Moreover, high temperatures induces finer droplet dispersion (Sidaoui, Sultan, and Qiu 2016).

The mechanical stirring is performed for 10 minutes after complete dispersion of oil phase into the mixture to ensure thorough solubilization. The mixture is then sonicated using a probe sonicator at a power of 300watts to 400 watts (with respect to the mixture volume) for 5 minutes. Sonication breaks the droplets and micelle phase further allowing for better homogenization and generation of a strong emulsion, the acoustic cavitation creates free radicals of different functional groups which provide better interaction of the Pickering particles with oil and water phases at micro level (Keramat, Kheynoor, and Golmakani 2022; Mason and Lorimer 2002). The emulsion appears mainly white with black polka dots which indicates the presence of dispersed activated carbon particles.

Stability

Emulsions were screened for optimum emulsifier concentration (1%, 3%, and 5%) at 0ppm, 10,000ppm, and 30,000ppm salinities under room temperature and 80°C (using the Binder, FD 115 oven) for 24 hours. The emulsion's stability was assessed by measuring the amount of phase separation (Hu et al. 2017). The

resulting measurement was used to determine the ESI using the following formula (Aryee, Agyei, and Udenigwe 2018; Chang 1994; Saikia et al. 2020):

$$ESI = \left[1 - \frac{V_o}{V_e} \right] * 100 \quad (1)$$

Where, V_o is the volume of oil separated from the emulsion and V_e is the volume of emulsion.

Rheology

To understand how the emulsions behave under thermal and mechanical (shear) stress, a rheological study was conducted using the Discovery Hybrid Rheometer (DHR 1). The concentric cylinder plate geometry was used which ensures accurate measurements by preventing slipping and evaporation of fluid under high temperature (Yousufi et al. 2023). Rotational rheology of the emulsion samples with varying emulsifier concentrations (1%, 3%, and 5%) at different salinities (0ppm and 30,000ppm) salinities were examined under varying shear rate ranging from 0.1 to 1000 1/sec and temperature 30°C, 60°C, and 90°C to observe the shear thinning and thickening behavior.

Contact Angle

To evaluate the effectiveness of a conformance agent it is necessary to analyze its interaction with the different saturated zones (oil and water saturated). For this purpose, three Berea sandstone rock pellets of 25mm × 10mm with a 1.5 mm thickness were used for contact angle measurements at 90°C. One pellet was not saturated and used dry as a reference, while the second pellet was saturated with crude oil and the third pellet was saturated with 30,000 ppm brine for 5 days. The contact angle measurements were made using the Rame-Hart goniometer, 260-U1, US. Using the sessile drop method contact angle measurements were made at the two temperatures. A small emulsion droplet was carefully placed on each sandstone sample using a precision syringe. The initial contact angle and droplet dimensions (width, height, area, and volume) were recorded immediately after placement and at regular intervals till complete adsorption to observe any changes over time. The contact angle measurements for each rock pellet were triplicated to achieve a consistent value thus the mean contact angles have been reported.

Core Flooding

The impact of emulsion slug size on plugging performance was assessed using a Basic Core Flooding System (BCFS- 350) from Polygon Synergy Ventures. A standard 3-inch long, 1.5-inch diameter core was employed for the experiment. Prior to core flooding, initial core porosity and permeability were measured using a Vinci Technologies POROPERM instrument, as detailed in Table 1. After characterization, the core was placed in the BCFS-350 system and set at a temperature of 50°C, a confining pressure of 1200 psi, and a back pressure of 6000 psi. The sample was then saturated with 30,000 ppm salinity brine, and initial permeability was determined using Darcy's Law by measuring flow rate and differential pressure at various flow rates (0.1, 0.25, and 0.4 cc/min).

Table 1—Berea Sandstone core sample properties

CoreSample	Diameter(mm)	Length(mm)	Weight(g)	Effective Porosity(%)	Pore Volume(cc)	Permeability(mD)
Berea Sandstone	38.09	76.17	169.411	31.4	27.26	705.259

Following brine injection, three successive 7PV emulsion slugs were injected into the core at a rate of 1 cc/min, with brine injection at different flow rates (0.1, 0.25, and 0.4 cc/min) conducted after each slug to evaluate return permeability. This process allowed for the assessment of how increasing emulsion volume within the core affects plugging performance.

Results and Discussion

Stability

The emulsions with different emulsifier concentrations (1%, 3%, and 5%) were evaluated for thermal stability under ambient and 90°C using the bottle test for a period of 24 hours. Figure 1a shows that all the emulsions under room temperature had no significant phase separation. The emulsion stability increases with increase in emulsifier concentration this is because with the increase in particle volume there are ample number of particles to adsorb at the oil-water interface, since the emulsifier is a porous composite of activated carbon and bentonite this provides dual steric stabilization as the droplets are not only surrounded by the Pickering particles, the activated carbon especially adsorbs the oil droplets within its pores thus providing a dual physical barrier along with the laced bentonite being hydrophilic showing similar affinity for the aqueous phase thereby providing a sound hydrophilic-lipophilic balance. Moreover, as the critical particle concentration increases these excess particles tend to function as spacers between the oil droplets thus developing a strong structural network which prevents droplet coalescence.

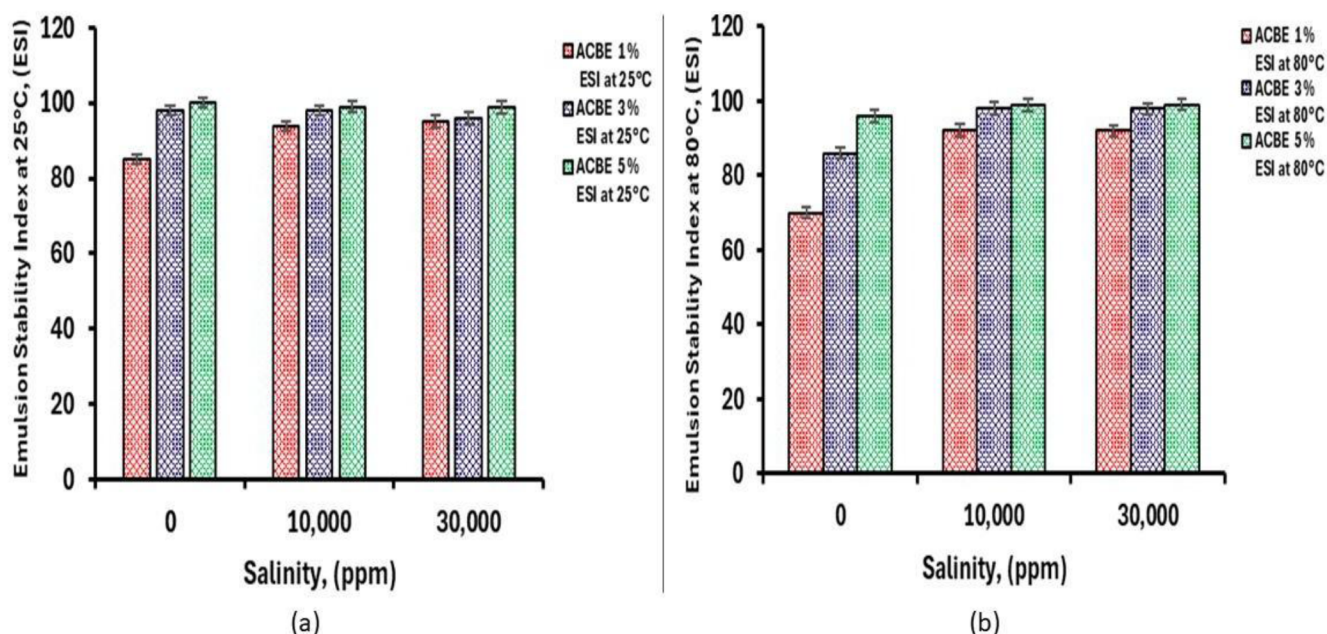


Figure 1—(a), (b) Effect of emulsifier concentration and salinity on emulsion stability. Emulsion Stability Index (ESI) values increased with both higher emulsifier concentrations and increasing salinity levels, demonstrating the strong hydrophilic-lipophilic balance provided by the porous nanocomposite emulsifier.

As the emulsifier is a combination of hydrophilic and lipophilic porous materials (bentonite coated on activated carbon) therefore with the increase in salinity (to 10,000 ppm, and 30,000 ppm) there is further enhancement of the particle structural network as the salt ions also seem to act as co-spacers to the particles providing additional electrostatic repulsion between the droplets (Hatchell, Song, and Daigle 2022; Du Sorbier, Aimable, and Pagnoux 2015). This is evident from the 85% being the lowest ESI of the emulsion with 1% emulsifier concentration and 0 ppm salinity while the ESI increased with increase in solid particle concentrations (emulsifier and salt ions) from 94%, 95% for 1% emulsifier concentration at 10,000 ppm and 30,000 ppm salinities respectively to 98% and 99% at 3% and 5% emulsifier concentration under varying salinities.

In comparison to the room temperature emulsion stability at 90°C (Figure 1b) shows similar behavior by retaining significant amount (the lowest values being 70%, and 86%) only occurring for the emulsions with 1% and 3% emulsifier concentration at 0 ppm salinity. While the rest of the emulsions with increased emulsifier concentration and salinity yield 92% ESI values for 1% emulsifier concentration-based emulsion

under both higher salinities. Moreover, 3% and 5% emulsifier concentration-based emulsions maintain 98% and 99% ESI values under 10,000ppm and 30,000ppm salinities, respectively.

The slight decrease in emulsion stability is due to increased kinetic energy (which tend to adversely affect the repulsive steric forces generated by the solid particles) and decreased water viscosity (which leads to coalescence) at elevated temperatures (90°C). However, the presence of excess solid particles (Pickering emulsifier and salt ions) acting as fortified particle structural network tends to mitigate these effects at higher temperature (90°C) by providing a strong steric hinderance between the droplets, thereby developing a resilient colloidal system which is able to withstand changes in temperature and salinity under static conditions.

Rheology

To further evaluate the emulsions resistance to thermal and mechanical (shear) fluctuations rotational rheological screening was conducted under varying temperature (30°C, 60°C, and 90°C), and shear stress (0.1 to 1000 1/sec). [Figure 2a](#) and [2b](#) provide the graphical representation of flow sweep rheology tests conducted at 0ppm and 30,000ppm, respectively. The rheological profiles concur with the bottle test analysis that the emulsifier concentration has significant impact upon emulsion stability. Emulsion viscosity critically influences colloidal stability as well as the mobility, which is highly affected during elevated temperatures. This is because the water viscosity decreases making it easier for the droplet to move around and interact with one another. However, the Brownian motion-based repulsion between the water molecules and oil droplets induced acts as a counter mechanism to prevent coalescence but to a limited extent due to larger size of the oil droplets especially when emulsifier concentration is limited ([Farah et al. 2005](#); [Piskunov et al. 2022](#)).

This is evident from 1% emulsifier concentration-based emulsions where firstly the higher temperature slightly reduces particle's effectiveness in adsorbing and surrounding individual droplets and due to lower concentration Pickering particles are unable to create a strong structural network because of which the smaller oil droplets migrate towards the larger droplets. As the particle (emulsifier) concentration increases (up to 3% and 5%) the rheological behavior improves, because there are ample particles for droplet adsorption, and coating. The increased number of particles, minimize droplet aggregation (flocking), detachment from the particle surface (desorption), and free movement.

The excess particles which similar to their function as spacers against coalescence under high shear stress act as bridging material. The structural particle network around the droplets allows for the droplets to deform and rearrange against deformation caused by the shear stress. Therefore, the samples with higher concentrations of Pickering particles exhibit greater resistance to viscosity reduction under high shear rates and temperature by retaining higher viscosity due to presence of higher solid mass concentration ([Mohsin Yousufi et al. 2025](#)).

The higher salinity (30,000ppm) rheological profiles ([Figure 2b](#)) provide similar behavior to 0ppm salinity profiles especially for the 1% emulsifier concentration-based emulsions. Emulsions with 3% emulsifier concentration have greater resistance to thermal and shear changes by retaining greater viscosity than emulsions under 0ppm salinity. This is because of the presence of salt ions acting as co-bridging materials to the Pickering particle-based structural network. This effect becomes more dominant at higher temperatures (60°C and 90°C) where although water viscosity decreases over time, the higher viscosity provided by the combination of salt ions working with the porous hydrophilic-lipophilic Pickering particles is able to mitigate thermal and shear destabilization.

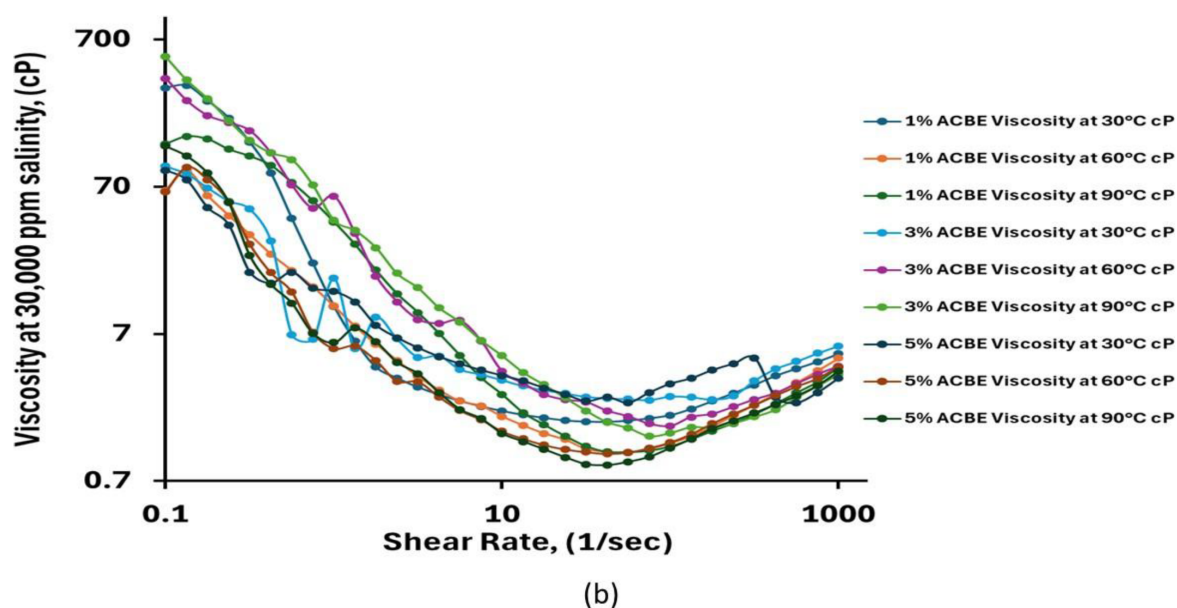
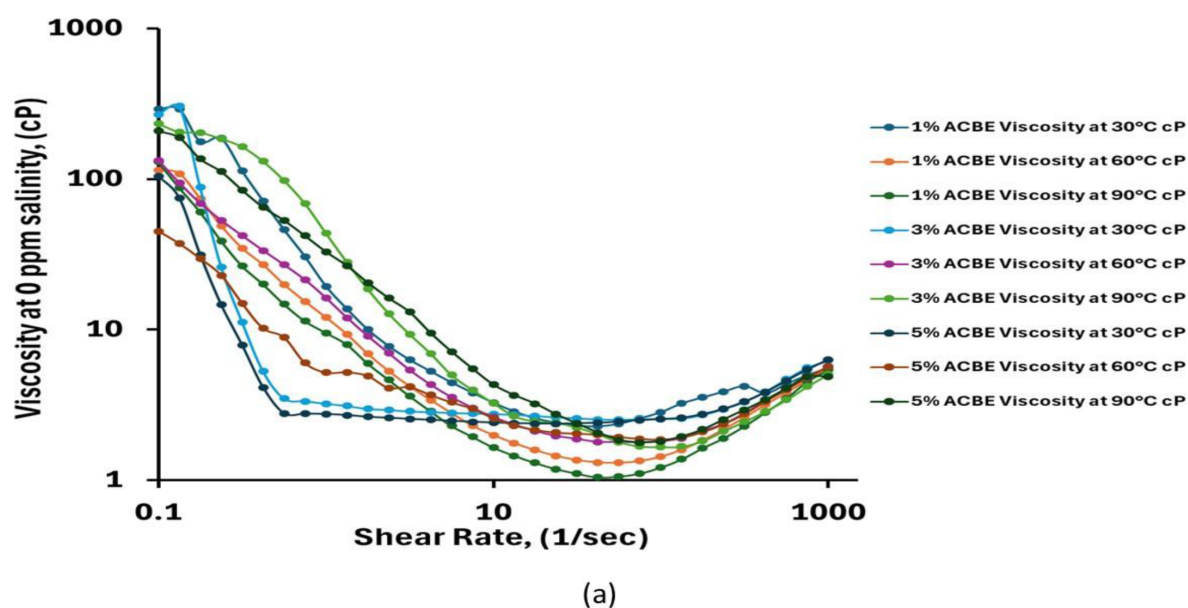


Figure 2—(a), (b) Effect of emulsifier concentration and salinity on emulsion rheological stability. Rheological profiles of emulsions subjected to varying temperatures (30°C, 60°C, and 90°C) and shear stresses (0.1-1000 1/s) at different emulsifier concentrations (1%, 3%, and 5%) and salinities (0 ppm and 30,000 ppm). The results demonstrate the enhanced resistance of emulsions with higher emulsifier concentrations and salinity to thermal and mechanical degradation, highlighting the significance of the porous nanocomposite emulsifier in maintaining emulsion stability and viscosity.

The 5% emulsifier concentration-based emulsions tend to yield lower viscosity profiles than the 3% concentration-based emulsions. Although the 5% concentration based- emulsions are able to maintain desired colloidal integrity against the shearing forces the 3% emulsifier concentration appears to be the better and more optimized concentration. The 5% emulsions particles show lagging behavior as these have become overcrowded, and the particles are interfering with each other. This concentration exceeds the optimized volume needed for the formation of a strong network, causing disruption among particle-stabilized droplets. As a result, the weak, overcrowded particle structural network will have limited mobility and reduced viscosity under high shear stress.

Contact Angle

Contact angle measurements provide a screening tool for evaluating the potential of emulsions as effective plugging agents to enhance reservoir sweep efficiency. High-permeability zones, subjected to extensive brine flooding, have been rendered water-wet, while low-permeability zones retain significant oil saturation. The contact angle data presented in Table 2 indicates the emulsion's suitability as a plugging agent. Given the contact angle of the dry sandstone rock pellet with the emulsion is 18° , the significantly larger contact angle between the oil-saturated rock pellet (69.5°) and the emulsion compared to the water-saturated rock pellet (26.3°) suggests a strong preferential wetting of the water-wet zones. This selective wetting behavior will drive the emulsion towards high-permeability zones, blocking them and diverting water flow towards lower-permeability, oil-bearing zones, thereby improving oil recovery.

Table 2—Wettability of emulsion droplets on oil- and brine-saturated sandstone pellets

Sample	Contact Angle	Temperature, °C
Sandstone	18	90
Oil Saturated sandstone	69.5	
Sandstone saturated in 30kppm brine	26.3	

Core Flooding

The initial return permeability after the first brine injection for saturating the core was determined. As mentioned in the methodology section the brine injection at different flow rates (0.1, 0.25, and 0.4 cc/min) was conducted for a period of 30 minutes per each flow rate. The initial brine saturation returns permeability of 666.169mD reduced the original core permeability by 5.5%. After which the core was injected with 7PV emulsion slug at a flow rate of 1cc/min for 15 minutes, the high-pressure buildup (64.7 psi) was evident due to the high emulsion viscosity as seen in the graphical illustration presented in Figure 3. The core was allowed to age for shut-in period of 24 hours in order to ensure that effective plugging is induced by the emulsion. After the aging period the return permeability was determined of 11.72mD showing a significant permeability reduction of 98.23% from the initial brine injection.

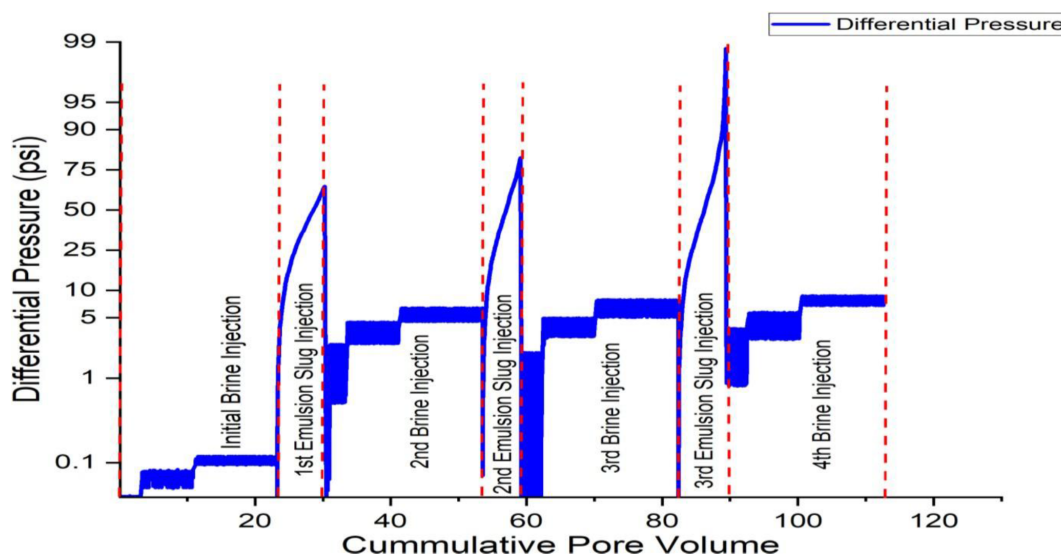


Figure 3—Coreflooding experiment demonstrating the effect of emulsion slug injections on differential pressure. The results show a significant increase in differential pressure upon the injection of each emulsion slug, suggesting effective plugging of the core. The gradual decrease in differential pressure during subsequent brine injections indicates the persistence of the plugging effect.

The core was further injected with a 2nd emulsion slug injection for the same volume, and flow rate. The build up pressure was higher than the 1st slug injection reaching up to 80 psi which also signifies the resilience of the plugging effect generated by 1st emulsion slug. After a 24-hour shut-in time the same process of measuring return permeability was observed which exhibited a small change in reduction of 8.94mD from the first slug injection and transitioning the cumulative permeability reduction to 98.65%. The third and final emulsion slug injection was performed under the same parameters during which higher differential pressure values were recorded as high as 97.03 psi this was followed by the shut-in time and return permeability measurement of 8.91mD progressing the permeability reduction to mere 98.66%.

Given the significant initial improvement in permeability reduction and the diminishing returns from additional injections, it's encouraging to see that a single emulsion slug injection was highly effective in achieving the desired conformance control objective. While further injections might offer marginal benefits, the substantial initial results suggest that the current strategy is a promising and cost-effective approach for improving reservoir performance.

Conclusion

The experimental results demonstrate that the synthesized emulsion, stabilized by nanocomposite Pickering particles, exhibits promising properties as a conformance agent for water shut-off. The emulsion displayed exceptional stability under various conditions, including elevated temperatures and varying salinities. This stability is attributed to the dual steric stabilization provided by the porous composite Pickering particles and the synergistic effect of salt ions. Additionally, the emulsion exhibited favorable rheological behavior, maintaining high viscosity under both elevated temperatures and shear stresses. This indicates the formation of a strong structural network by the Pickering particles and the bridging effect of salt ions. Furthermore, contact angle measurements confirmed the emulsion's preference for wetting water-wet surfaces, suggesting its potential to selectively plug high-permeability zones in reservoirs. The core flooding experiments further validated the emulsion's effectiveness as a plugging agent, demonstrating a significant reduction in permeability after a single slug injection. Subsequent injections provided diminishing returns, indicating that the initial plugging was highly effective. Overall, these findings suggest that the synthesized emulsion has the potential to be a valuable tool for improving reservoir conformance and enhancing oil recovery.

To enhance the application of this emulsion as a conformance agent, future research should evaluate its performance in cores with varying degrees of heterogeneity, as Berea sandstone represents a relatively homogeneous material. A comprehensive analysis of the emulsion's interfacial properties and droplet size distribution is essential to understand their influence on plugging efficiency. Determining the optimal emulsion slug volume for effective plugging in diverse heterogeneous cores will contribute to process optimization. Computational upscaling of these coreflooding results can facilitate the application of the emulsion to Malaysian reservoirs, potentially mitigating excess water production and consumption, thereby reducing operational costs and increasing oil recovery from untapped low- permeability zones.

References

- Abu Zaid, Nansee S. K., Mustafa S. Nasser, Sagheer A. Onaizi, Nansee S K Abu Zaid, Mustafa S. Nasser, and Sagheer A. Onaizi. 2024. "Pickering Emulsions Stabilized by Metal-Organic Frameworks, Graphene-Based Materials, and Carbon Nanotubes: A Comprehensive Review." *Journal of Molecular Liquids* **393** (July 2023): 123617. <https://doi.org/10.1016/j.molliq.2023.123617>.
- Adewunmi, Ahmad A., Theis Solling, Abdullah S. Sultan, and Tinku Saikia. 2022. "Emulsified Acid Systems for Oil Well Stimulation: A Review." *Journal of Petroleum Science and Engineering* **208**: 109569. <https://doi.org/10.1016/j.petrol.2021.109569>.
- Aryee, A.N.A., D Agyei, and C.C. Udenigwe. 2018. "Impact of Processing on the Chemistry and Functionality of Food Proteins." In *Proteins in Food Processing*, edited by Rickey Y B T - *Proteins in Food Processing* (Second Edition) Yada, 27-45. Elsevier. <https://doi.org/10.1016/B978-0-08-100722-8.00003-6>.

- Bai, Baojun, Ming Han, Yuxiang Li, Mingzhen Wei, Y Gao, and Jean Paul Coste. 2000. "Selective Water Shutoff Technology Study and Application of W/O Emulsions." In SPE/DOE Improved Oil Recovery Symposium. OnePetro.
- Chang, Pahn Shick. 1994. "Determination of Emulsion Stability Index in W/O Emulsion." *Analytical Science and Technology* 7 (2): 233-36.
- Farah, Marco A, Roberto C Oliveira, Jorge Navaes Caldas, and Krishnaswamy Rajagopal. 2005. "Viscosity of Water-in-Oil Emulsions: Variation with Temperature and Water Volume Fraction." *Journal of Petroleum Science and Engineering* 48 (3): 169-84. <https://doi.org/https://doi.org/10.1016/j.petrol.2005.06.014>.
- Fink, Johannes. 2021. "Chapter 16 - Enhanced Oil Recovery." In *Petroleum Engineer's Guide to Oil Field Chemicals and Fluids* (Third Edition), 643-731. Gulf Professional Publishing. <https://doi.org/https://doi.org/10.1016/B978-0-323-85438-2.00016-5>.
- Frijters, Stefan, Florian Gunther, and Jens Harting. 2012. "Effects of Nanoparticles and Surfactant on Droplets in Shear Flow." *Soft Matter* 8 (24): 6542-56. <https://doi.org/10.1039/c2sm25209k>.
- Fu, Lipei, Qianli Ma, Kaili Liao, Junnan An, Jinmei Bai, and Yanfeng He. 2022. "Application of Pickering Emulsion in Oil Drilling and Production." *Nanotechnology Reviews* 11 (1): 26-39. <https://doi.org/10.1515/ntrev-2022-0003>.
- Hagar, Haithm Salah, Shiferaw Regassa Jufar, Jang Hyun Lee, Najib Al-mahbashi, M. Basheer Alameen, Sunil Kwon, Ahmad Hussaini Jagaba, and Upaka Rathnayake. 2023. "Chitin Nanocrystals: A Promising Alternative to Synthetic Surfactants for Stabilizing Oil-in-Water Emulsions." *Case Studies in Chemical and Environmental Engineering* 8 (September): 100503. <https://doi.org/10.1016/j.csee.2023.100503>.
- Hatchell, Daniel, Wen Song, and Hugh Daigle. 2022. "Examining the Role of Salinity on the Dynamic Stability of Pickering Emulsions." *Journal of Colloid and Interface Science* 608: 2321-29. <https://doi.org/10.1016/j.jcis.2021.10.154>.
- Hossain, Kazi M. Zakir, Laura Deeming, and Karen J. Edler. 2021. "Recent Progress in Pickering Emulsions Stabilised by Bioderived Particles." *RSC Advances*. <https://doi.org/10.1039/d1ra08086e>.
- Hu, Yin Ting, Yuwen Ting, Jing Yu Hu, and Shu Chen Hsieh. 2017. "Techniques and Methods to Study Functional Characteristics of Emulsion Systems." *Journal of Food and Drug Analysis* 25 (1): 16-26. <https://doi.org/10.1016/j.jfda.2016.10.021>.
- IEA World Energy Outlook. 2019. "World Energy Outlook." IEA, Paris License: CC BY 4.0. 2019. <https://www.iea.org/reports/world-energy-outlook-2019>.
- Keramat, Malihe, Najme Kheynoor, and Mohammad-taghi Golmakani. 2022. "Oxidative Stability of Pickering Emulsions." *Food Chemistry: X* 14 (March): 100279. <https://doi.org/10.1016/j.fochx.2022.100279>.
- Khan, Mohammed Yahaya, Z A Abdul Karim, A Rashid A Aziz, and Isa M Tan. 2014. "Experimental Investigation of Microexplosion Occurrence in Water in Diesel Emulsion Droplets during the Leidenfrost Effect," 7079-84.
- Kim, I., A. J. Worthen, M. Lotfollahi, K. P. Johnston, D. A. Dicarolo, and C. Huh. 2017. "Nanoparticle-Stabilized Emulsions for Improved Mobility Control for Adverse-Mobility Waterflooding." IOR NORWAY 2017 - 19th European Symposium on Improved Oil Recovery: Sustainable IOR in a Low Oil Price World. <https://doi.org/10.3997/2214-4609.201700314>.
- Li, Shidong, Long Yu, Hon Chung Lau, and Ludger P. Stubbs. 2020. "Experimental Study of Mobility Control by Clay-Stabilized Pickering Emulsions in High-Salinity Reservoirs." In Proceedings - SPE Annual Technical Conference and Exhibition, 1-15. <https://doi.org/10.2118/201256-MS>.
- Li, Wei, Bo Jiao, Sisheng Li, Shah Faisal, Aimin Shi, Weiming Fu, Yiyang Chen, and Qiang Wang. 2022. "Recent Advances on Pickering Emulsions Stabilized by Diverse Edible Particles: Stability Mechanism and Applications." *Frontiers in Nutrition* 9 (May): 1-17. <https://www.frontiersin.org/articles/10.3389/fnut.2022.864943>.
- Mandal, Ajay, Achinta Bera, Keka Ojha, and T Kumar. 2012. "Characterization of Surfactant Stabilized Nanoemulsion and Its Use in Enhanced Oil Recovery." *SPE Int. Oilfield Nanotechnol. Conf.* 6 (7): 80-92. <https://doi.org/10.2118/155406-ms>.
- Mason, T J, and J P Lorimer. 2002. "Applied Sonochemistry: Uses of Power Ultrasound In." *Chemistry*.
- McAuliffe, C D. 1973. "Crude Oil-in-Water Emulsions to Improve Fluid Flow in an Oil Reservoir." *J. Pet. Technol.*, 721.
- Mohsin Yousufi, Muhammad, Iskandar bin Dzulkarnain, Mysara Eissa Mohyaldinn Elhaj, Suriati binti Sufian, Berihun Mamo Negash, Haithm Salah Hagar, and Shehzad Ahmed. 2025. "Evaluation of Activated Carbon as a Pickering Emulsion Stabilizer for Conformance Control at High Temperature and Salinity: A Focus on Stability and Rheology." *Journal of Molecular Liquids* 419 (December 2024): 126764. <https://doi.org/10.1016/j.molliq.2024.126764>.
- Pandey, Aseem, Ali Telmadarreie, Milana Trifkovic, and Steven Bryant. 2018. "Cellulose Nanocrystal Stabilized Emulsions for Conformance Control and Fluid Diversion in Porous Media." In Proceedings - SPE Annual Technical Conference and Exhibition, 2018-Sept:24-26. <https://doi.org/10.2118/191609-ms>.
- Patel, Arvindbhai, Anil Kumar Singh, Nikhil Bidwai, Sakshi Indulkar, and Vivek Gupta. 2021. "Innovative Surfactant Chemistry Offers the Performance Advantages to Invert Emulsion Drilling Fluids While Drilling Under Challenging Environments." SPE International Conference on Oilfield Chemistry. <https://doi.org/10.2118/204362-MS>.

- Piskunov, Maxim, Alexander Ashikhmin, Nikita Khomutov, and Alexandra Semyonova. 2022. "Effects of Wall Temperature and Temperature-Dependent Viscosity on Maximum Spreading of Water-in-Oil Emulsion Droplet." *International Journal of Heat and Mass Transfer* **185**: 122442. <https://doi.org/https://doi.org/10.1016/j.ijheatmasstransfer.2021.122442>.
- Saafie, Nabilah, Nonni Soraya Sambudi, Mohd Dzul Hakim Wirzal, and Suriati Sufian. 2023. "Effect of Hydrothermal Conditions on Kenaf-Based Carbon Quantum Dots Properties and Photocatalytic Degradation." *Separations* **10** (2). <https://doi.org/10.3390/separations10020137>.
- Saikia, Tinku, Abdullah Sultan, Assad A. Barri, Nur Iman Khamidy, Abdulmalek A. Shamsan, Ayman Almohsin, and Mohammed Bataweel. 2020. "Development of Pickering Emulsified Polymeric Gel System for Conformance Control in High Temperature Reservoirs." *Journal of Petroleum Science and Engineering* **184** (January): 106596. <https://doi.org/10.1016/j.petrol.2019.106596>.
- Sarkar, Anwesha, Shuning Zhang, Melvin Holmes, and Rammile Ettelaie. 2019. "Colloidal Aspects of Digestion of Pickering Emulsions: Experiments and Theoretical Models of Lipid Digestion Kinetics." *Advances in Colloid and Interface Science* **263**: 195-211. <https://doi.org/https://doi.org/10.1016/j.cis.2018.10.002>.
- Shi, Yunlei, Dazhen Xiong, Zhiyong Li, Huiyong Wang, Jikuan Qiu, Hucheng Zhang, and Jianji Wang. 2020. "Ambient CO₂/N₂ Switchable Pickering Emulsion Emulsified by TETA-Functionalized Metal-Organic Frameworks." *ACS Applied Materials and Interfaces* **12** (47): 53385-93. <https://doi.org/10.1021/acsami.0c13157>.
- Sidaoui, Ziad, Abdullah S Sultan, and Xiangdong Qiu. 2016. "Viscoelastic Properties of Novel Emulsified Acid Using Waste Oil : Effect of Emulsifier Concentration, Mixing Speed and Temperature." In SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition, SPE-182845-MS, 1-11. Dammam, Saudi Arabia: Society of Petroleum Engineers. <https://doi.org/10.2118/182845-MS>.
- Sorbier, Q Monegier Du, A Aimable, and C Pagnoux. 2015. "Influence of the Electrostatic Interactions in a Pickering Emulsion Polymerization for the Synthesis of Silica-Polystyrene Hybrid Nanoparticles." *Journal of Colloid and Interface Science* **448**: 306-14.
- Sun, Ziqi, Wanfen Pu, Renbao Zhao, and Shishi Pang. 2022. "Study on the Mechanism of W/O Emulsion Flooding to Enhance Oil Recovery for Heavy Oil Reservoir." *Journal of Petroleum Science and Engineering* **209** (May 2021): 109899. <https://doi.org/10.1016/j.petrol.2021.109899>.
- Wang, Wuchao, Xiaohu Dong, Huiqing Liu, Yan Peng, Zhangxin Chen, Yu Li, and Yunfei Guo. 2022. "Fly Ash Nanoparticle-Stabilized Emulsions for Improve Mobility Control Application." Society of Petroleum Engineers - SPE EuroPEC - Europe Energy Conference Featured at the 83rd EAGE Annual Conference and Exhibition, EURO 2022. <https://doi.org/10.2118/209646-MS>.
- Wu, Ke Jun, Edmund C.M. Tse, Congxiao Shang, and Zhengxiao Guo. 2022. "Nucleation and Growth in Solution Synthesis of Nanostructures - From Fundamentals to Advanced Applications." *Progress in Materials Science* **123** (August 2020): 100821. <https://doi.org/10.1016/j.pmatsci.2021.100821>.
- Xu, Ke, Peixi Zhu, Tatiana Colon, Chun Huh, and Matthew Balhoff. 2017. "A Microfluidic Investigation of the Synergistic Effect of Nanoparticles and Surfactants in Macro-Emulsion-Based Enhanced Oil Recovery." *SPE Journal* **22** (2): 459-69. <https://doi.org/10.2118/179691-PA>.
- Yang, Yunqi, Zhiwei Fang, Xuan Chen, Weiwang Zhang, Yangmei Xie, Yinghui Chen, Zhenguo Liu, and Weien Yuan. 2017. "An Overview of Pickering Emulsions: Solid-Particle Materials, Classification, Morphology, and Applications." *Frontiers in Pharmacology* **8** (MAY): 1-20. <https://doi.org/10.3389/fphar.2017.00287>.
- Yousufi, Muhammad Mohsin, Iskandar Dzulkarnain, Mysara Eissa, Mohyaldinn Elhaj, and Shehzad Ahmed. 2023. "A Perspective on the Prospect of Pickering Emulsion in Reservoir Conformance Control with Insight into the Influential Parameters and Characterization Techniques." *Processes* **11** (2672): 1-29. <https://doi.org/10.3390/pr11092672>.
- Zeng, Zhifeng, Fei Ma, Shuangjia Wang, Jiali Wen, Xiangqiong Jiang, Gaoyu Li, Yan Tong, Xiaolong Liu, and Jiuxing Jiang. 2024. "Zeolitic Pickering Emulsifier with Intrinsic Amphiphilicity." *Journal of the American Chemical Society* **146** (14): 9851-59. <https://doi.org/10.1021/jacs.3c14712>.
- Zhang, Xiao, Jia Nan Deng, Kai Yang, Qian Li, Sen Yao Meng, Xiu Xia Sun, Zhao Zheng Song, et al 2022. "High-Strength and Self-Degradable Sodium Alginate/Polyacrylamide Preformed Particle Gels for Conformance Control to Enhance Oil Recovery." *Petroleum Science*, no. xxxx. <https://doi.org/10.1016/j.petsci.2022.06.012>.
- Zhang, Yuanxia, Yan Bao, Wenbo Zhang, and Ru Xiang. 2023. "Factors That Affect Pickering Emulsions Stabilized by Mesoporous Hollow Silica Microspheres." *Journal of Colloid and Interface Science* **633**: 1012-21. <https://doi.org/10.1016/j.jcis.2022.12.009>.